MA411 Notes 5

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12 Load balancing

Suppose you run a collection of N servers. At each time step, with probability $\lambda < 1$ a customer appears. They choose a server (somehow!) and add one job to its queue. Then, one server is chosen uniformly at random, and (if it has at least one job in its queue!) one job is completed and removed from the queue.

This is a standard model for Internet services. A (successful) service provider will buy sufficient servers—make N large enough—so that they are on average serving customers faster than customers arrive ($\lambda < 1$ in the model), because otherwise their servers are overloaded and crash.

How should customers choose the server? This is usually decided by the service provider (the customer doesn't really get to choose, but let's still imagine they are choosing for convenience of language).

The two most obvious choices are: (1) The customer picks the server with the shortest queue, and (2) The customer picks a random server.

(1) is bad for the service provider. Every time step, every server is interrupted and asked for its queue length, or alternatively the service provider needs to keep a list up to date of the queues—that would need to be on a separate MasterServer and in serious applications (Amazon, Google et cetera) the MasterServer would be overloaded trying to handle the flood of incoming traffic, even just needing to direct each customer to its own server.

Let's analyse (2). The 'right' choice is to count, for each $i \geq 0$, the number of servers with a queue of exactly i. Let's slightly modify the process (we'll see this is in practice irrelevant!): if a queue reaches length $N+1$ then instantly the customer waiting longest gives up and the queue remains of length N.

Example 1. *Consider the following process on $\mathbb{Z}_{\geq 0}^N$. We have N servers; at each time $t \in \mathbb{N}$, sample uniformly with replacement a pair (a, b) of servers; with probability $\lambda < 1$ add one to the queue of a (i.e. the entry number a); if this results in a queue of length $N+1$ then set the queue length back to N. Then, if the queue of b has positive length, subtract one from it.*

Let $\mathbf{Z}_n \in \{0, \ldots, N\}^N$ count the number of servers at time n whose queue has length at least i, for each $1 \leq i \leq N$. Let $(\mathcal{F}_n)_{n \in \mathbb{N}}$ be the natural filtration of the process.

In theory we should specify which process we actually mean here! The 'original' process is given by the random pair and the Bernoulli random variable with probability λ, while $(\mathbf{Z}_n)_{n \geq 0}$ has rather less information. But it will turn out not to matter here.

We need to evaluate $\mathbb{E}\big(\mathbf{Z}_n - \mathbf{Z}_{n-1}|\mathcal{F}_{n-1}\big)$. Consider the j component of this vector, for some $j \geq 1$ (i.e. counting the number of servers with queue length at least j).

We change the j component in two ways. If a is a server whose queue length is exactly $j-1$, then with probability λ we increment the j component. If b is a server whose queue length is exactly j, we decrement the j component. We get, for $j = 1$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(1)} - \mathbf{Z}_{n-1}^{(1)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(1)}-\mathbf{Z}_{n-1}^{(2)}}{N}\,,$$

for $1 < j < N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(j)} - \mathbf{Z}_{n-1}^{(j)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(j)}-\mathbf{Z}_{n-1}^{(j+1)}}{N}\,,$$

and for $j = N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(N)} - \mathbf{Z}_{n-1}^{(N)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{\mathbf{Z}_{n-1}^{(N-1)}-\mathbf{Z}_{n-1}^{(N)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(N)}}{N}\,.$$

where the superscript indicates component of the vector. The associated ODE is then

$$\frac{\mathrm{d}\mathbf{x}^{(1)}}{\mathrm{d}t}(t) = \lambda + \mathbf{x}^{(2)}(t) - (1+\lambda)\mathbf{x}^{(1)}(t)\,,$$

for $1 < j < N$

$$\frac{\mathrm{d}\mathbf{x}^{(j)}}{\mathrm{d}t}(t) = \lambda\mathbf{x}^{(j-1)}(t) + \mathbf{x}^{(j+1)}(t) - (1+\lambda)\mathbf{x}^{(j)}(t)\,,$$

and for $j = N$

$$\frac{\mathrm{d}\mathbf{x}^{(N)}}{\mathrm{d}t}(t) = \lambda\mathbf{x}^{(j-1)}(t) - (1+\lambda)\mathbf{x}^{(N)}(t)\,.$$

The purpose of our 'customer gives up' modification was to ensure our ODE system has finite dimension. But it's also realistic that customers who have to wait too long will give up.

This ODE is not a particularly nice one, but it is also not too hard to find out some facts.

For any initial state with all components non-negative, all components stay non-negative for all time: indeed $\mathbf{x}^{(j)}$ is always bounded below by e^{-t} times its initial value. To see this, observe that (provided all components are non-negative) we have $\mathbf{x}'(t) \geq -\mathbf{x}(t)$ in each component.

For any initial state satisfying $\mathbf{x}^{(j)}(t) \leq \lambda^j$ for all $1 \leq j \leq N$ at $t = 0$, the same inequalities hold for all $t \geq 0$. As long as these inequalities hold in all components, for any j (these bounds also work for $j = 1$ and $j = N$..!) we have

$$\frac{\mathrm{d}\mathbf{x}^{(j)}}{\mathrm{d}t}(t) \leq \lambda \cdot \lambda^{j-1} + \lambda^{j+1} - (1+\lambda)\mathbf{x}^{(j)}(t)$$

and substituting $y = \lambda^j - \mathbf{x}^{(j)}(t)$ this becomes

$$-y' \leq (1+\lambda)y$$

so $y' \geq -(1+\lambda)y$, so $y \geq y_0 e^{-(1+\lambda)t}$ does not become negative.

It's not too hard to see that the differential equation solution actually converges to a point which is quite close to $(\lambda, \lambda^2, \dots)$. Indeed, if we plug in those values, we obtain a

derivative of zero for all but the final ($j = N$) differential equations. The 'real' fixed point will have all coordinates very slightly smaller than $(\lambda, \lambda^2, \dots)$.

What we easily can prove, though, is that (assuming, as we justified, all coordinates are non-negative) the fixed point has jth coordinate at least $\left(\frac{\lambda}{1+\lambda}\right)^j$. This is what we would get from the jth ODE equation by substituting $x^{(j-1)} \geq \left(\frac{\lambda}{1+\lambda}\right)^{j-1}$ and $x^{(j+1)} \geq 0$; we therefore get this lower bound by induction. The point of this is that it justifies that our fixed point has coordinates which decrease exponentially in j (even if we don't know exactly what the base of the exponent is).

We want to apply the Differential Equations Theorem from the previous notes to argue that the random process will track this ODE. We can take $d = N$ and $\mathcal{D} = \mathbb{R} \times [-2, 2]^N$, and $B = 1$. Our function is easily seen to be 4-Lipschitz continuous since $\lambda < 1$ (this holds even on all of $\mathbb{R}^{N+1}$) and we can take $R = 8$ (this is just taking the maximum absolute value of each term and plugging in 2 as the maximum absolute value of each component of $\mathbf{x}$).

Since the ODE solution, from (for example; we could take any other initial state with the jth component in $[0, \lambda^j]$ for each j) the all zeroes initial state, stays within $[0, 1]^N$, we can choose τ to solve for t

$$2RN^{-1/4}e^{8Lt} = N^{-0.2}$$

which gives $\tau = \frac{1}{8L} \log\left(\frac{1}{2R}N^{0.05}\right)$; this is something like $\log N$.

We conclude that with probability tending to 1 as $N \to \infty$ (in fact, exponentially fast!) we have for all $n \leq \tau N$:

$$\left\|\mathbf{Z}_n - N\mathbf{x}\left(\tfrac{n}{N}\right)\right\|_\infty \leq 2N^{3/4}Re^{8Ln/N}\,.$$

What does this mean? If you are Google, you want to avoid too many customers being in long queues: you depend on your service having a reputation for being fast. Let's (somewhat arbitrarily) say we want to know the queue length for the worst $N^{0.8}$ servers. Recall that there is a scale factor N between the ODE solution and the process, so we're trying to solve for j (even in the best case): $\left(\frac{\lambda}{1+\lambda}\right)^j = 2N^{-0.2}$; we will need to take j something like $\log N$.

Realistically, we're actually solving (roughly) $\lambda^j = N^{-0.2}$, so $j = 0.2\frac{\log N}{\log \lambda^{-1}}$. Either way, this is not good for Google; customers want instant service—and quite a number of these customers will sit waiting something like $\log N$ times as long as they expect.

Something which is also bad is that only about λN servers are actually working. The rest have been paid for and are using electricity, but not actually doing any jobs. (Of course, which servers are idle will change constantly, but the fraction stays about the same.) And you cannot simply choose N to make λ only slightly less than 1, because then $\log \lambda^{-1}$ gets very small and your customer wait times get rapidly worse. For a data point: Google currently operates something like 10^6 servers (within a factor 10, say).

This last comment is 'external to the model'. Our model simply assumes that N and λ are given to us and we have to analyse the stochastic model.

However, if you are Google, while you don't really control the rate at which customers arrive, you do control the number N of servers you are operating; you can always choose to

buy more. If you observe the number of customer requests goes up, so that customers arrive more often, you'll buy more servers—increase N. In our scaling, N units of time is about how long we expect to wait for a single server to complete a job (because we model the job completion as one random server every time unit), so if we double N, that corresponds to the time unit of our model being half as long in real time (of course when we buy twice as many servers, it doesn't make the real time to complete a job double, that stays the same!). If the customer demand doesn't change, that would correspond to halving λ. This is what's meant by saying Google can in effect choose λ.

Here is what is actually done: each customer chooses *two* uniform random servers, and joins the shorter of the two queues. (In fact, a routing server makes the choice for the customer! but the point is that this choice doesn't need the routing server to maintain a list of queues, and there can be (and are!) multiple routing servers working in parallel; it doesn't change the model.)

Example 2. *Consider the following process on $\mathbb{Z}_{\geq 0}^N$. We have N servers; at each time $t \in \mathbb{N}$, sample uniformly with replacement a triple (a, b, c) of servers. If the queue of b is at least as long as that of a, with probability $\lambda < 1$ add one to the queue of a; if the queue of b is shorter than that of a, with probability $\lambda < 1$ add one to the queue of b. If this results in a queue of length $N + 1$ then set the queue length back to N. Then, if the queue of c has positive length, subtract one from it.*

Let $\mathbf{Z}_n \in \{0, \ldots, N\}^N$ count the number of servers at time n whose queue has length at least i, for each $1 \leq i \leq N$. Let $(\mathcal{F}_n)_{n\in\mathbb{N}}$ be the natural filtration of the process.

Again, we need to evaluate $\mathbb{E}\big(\mathbf{Z}_n - \mathbf{Z}_{n-1}|\mathcal{F}_{n-1}\big)$. Consider the j component of this vector, for some $j \geq 1$.

The analysis is rather similar to the above: indeed, the expectation of decrementing the j component is the same. However now there are two ways we can increment j: either a is a server with queue of length exactly $j-1$, and b of length at least $j-1$, or b is a server with queue of length exactly $j-1$ and a of length at least j. These two events are disjoint, so we can simply sum the conditional expectations (we're not double counting anything!). We get, for $j = 1$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(1)} - \mathbf{Z}_{n-1}^{(1)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N}\tfrac{N}{N} + \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(1)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(1)}-\mathbf{Z}_{n-1}^{(2)}}{N},$$

for $1 < j < N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(j)} - \mathbf{Z}_{n-1}^{(j)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(j-1)}}{N} + \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(j)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(j)}-\mathbf{Z}_{n-1}^{(j+1)}}{N},$$

and for $j = N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(N)} - \mathbf{Z}_{n-1}^{(N)}|\mathcal{F}_{n-1}\big) = \lambda\tfrac{\mathbf{Z}_{n-1}^{(N-1)}-\mathbf{Z}_{n-1}^{(N)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(N-1)}}{N} + \lambda\tfrac{\mathbf{Z}_{n-1}^{(N-1)}-\mathbf{Z}_{n-1}^{(N)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(N)}}{N} - \tfrac{\mathbf{Z}_{n-1}^{(N)}}{N}.$$

where the superscript indicates component of the vector. The associated ODE is then

$$\frac{\mathrm{d}\mathbf{x}^{(1)}}{\mathrm{d}t}(t) = \lambda + \mathbf{x}^{(2)}(t) - \mathbf{x}^{(1)}(t) - \lambda\big(\mathbf{x}^{(1)}(t)\big)^2,$$

for $1 < j < N$

$$\frac{\mathrm{d}\mathbf{x}^{(j)}}{\mathrm{d}t}(t) = \lambda\big(\mathbf{x}^{(j-1)}(t)^2 - \mathbf{x}^{(j)}(t)^2\big) - \mathbf{x}^{(j)}(t) + \mathbf{x}^{(j+1)}(t)\,,$$

and for $j = N$

$$\frac{\mathrm{d}\mathbf{x}^{(N)}}{\mathrm{d}t}(t) = \lambda\big(\mathbf{x}^{(N-1)}(t)^2 - \mathbf{x}^{(N)}(t)^2\big) - \mathbf{x}^{(N)}(t)\,.$$

This set of differential equations is somewhat harder to work with. However we can do some things. Provided the components of $\mathbf{x}(0)$ are monotone decreasing (which they clearly will be! we cannot start with more servers having a queue of length at least 5 than at least 4..!) then the coordinates of $\mathbf{x}(t)$ remain monotone decreasing for all t, and (having proved this) it is easy to check that if the coordinates additionally start non-negative, they will remain so.

We can guess (and it's not all that hard to prove) that this ODE will converge to a fixed point. As we did before, let's just prove bounds on the coordinates.

Suppose the system starts with $0 \le \mathbf{x}^{(j)}(t) \le \lambda^{2^j-1}$ for all j at $t = 0$. Then this collection of inequalities remains true for all t.

As before, we just plug in the bounds, noting that we obtain valid bounds for $j = 1$ and $j = N$ as well as intermediate j:

$$\begin{aligned}\frac{\mathrm{d}\mathbf{x}^{(j)}}{\mathrm{d}t}(t) &\le \lambda\big(\lambda^{2(2^{j-1}-1)} - \mathbf{x}^{(j)}(t)^2\big) - \mathbf{x}^{(j)}(t) + \lambda^{2^{j+1}-1} \\ &= \lambda^{2^j-1} + \lambda^{2^{j+1}-1} - \lambda\mathbf{x}^{(j)}(t)^2 - \mathbf{x}^{(j)}(t)\end{aligned}$$

and substituting $y = \lambda^{2^j-1} - \mathbf{x}^{(j)}(t)$ we get

$$y'(t) = -2\lambda^{2^j-1}y + \lambda y^2 - y$$

so that y cannot become negative as required.

Exercise 1. *Complete the application of the Differential Equations Theorem 18 by finding a suitable $\mathcal{D}$, and values for L, R, B and τ.*

Your choice of τ should ensure that $\big\|\mathbf{Z}_n - N\mathbf{x}\big(\frac{n}{N}\big)\big\|_\infty \le N^{0.8}$ holds for all $0 \le n \le \tau N$.

Solution to Exercise 1:

The value of B is still 1 as in the first example.

It's always best to first begin trying to compute L and R rather than immediately guess what $\mathcal{D}$ should be (we can see what regions might be 'bad' and we should try to avoid including in $\mathcal{D}$). But in this case, there aren't any particularly bad constants around. Each of our differential equations has at most four terms, with coefficients at most 1. We definitely do want to avoid the terms themselves being very large, but for this we can afford to look at the same $\mathcal{D} = \mathbb{R} \times [-2, 2]^N$ as for the first example; we quickly compute $L = R = 4 \cdot 2^2 = 16$ works on this domain. We could have been more careful and optimised this to get a smaller constant, but 16 isn't 'bad' given we obviously can't do better than say 1.

So what remains is to check on τ. Starting from $\mathbf{x}(0) = \mathbf{0}$, we argued that the jth coordinate of the ODE trajectory will remain between 0 and $\lambda^{2^j-1} \le 1$ for all time, so we

don't have to worry about the ball leaving $\mathcal{D}$ until the ball radius reaches 1. Since we are required to stop when the ball radius is $N^{-0.2}$ (corresponding to the process being off from the scaled ODE solution by $N \times N^{-0.2} = N^{0.8}$), we have to solve

$$2N^{3/4} \cdot 16 \cdot e^{8 \cdot 16\tau} = N^{0.8}$$

which is easy enough: rearrange and take logs to get

$$\tau = \tfrac{1}{128} \log\left(\tfrac{1}{32} N^{0.05}\right).$$

Let's now evaluate what this means for Google. We still have only about λN active servers at any given time, so if λ is significantly less than 1, this is still not good for Google.

However, we're now trying to solve a different equation to determine what the queue length is for the worst $N^{0.8}$ servers:

$$\lambda^{2^j-1} = N^{-0.2} \quad \text{so} \quad (2^j - 1)\log\lambda^{-1} = 0.2\log N \quad \text{so} \quad j = \tfrac{1}{\log 2}\log\left(1 + \tfrac{0.2\log N}{\log\lambda^{-1}}\right).$$

Recall that in the previous example, the queue length on the worst $N^{0.8}$ servers was $\frac{0.2\log N}{\log\lambda^{-1}}$; if we wanted to take λ close to 1, so that (for example) 99% of the servers are active, then that would mean (with realistic numbers!) that the queue length we previously got would be something like 200. This time, the load has dropped by taking an extra log: with the same λ, we get a queue length of a bit less than 8.

Leaving N in as a parameter, the difference here is between scaling as $\log N$, versus as $\log\log N$; there is a huge difference in how fast these functions grow.

Exercise 2. *Can Google do even better if customers are allowed to query* 3 *random servers, or generally* s *random servers? Write down the corresponding process, set up the associated ODE, and try to find out. You may find it useful to simulate the process and / or to use a computer to help find the fixed point of the ODE.*

Solution to Exercise 2:

Let's explicitly do this for 3 choices, and hope to see a pattern. The conditional expectation for decreasing queue length will be the same as for 1 and 2 choices; so what we need to do is, when we choose 3 uniform random servers a, b, c (with replacement, as before), calculate the chance that the one with the shortest queue will have queue length exactly $j-1$, and so contribute to $Z^{(j)}$ increasing. As before, we'll tie-break deterministically. We choose a if the other two queues are at least as long as a's queue, if we don't choose a then we choose b if its queue is at most as long as c's, and we choose c exactly when a and b have strictly longer queues. As before, this means we can easily enough write down the conditional expectations.

We get, for $j = 1$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(1)} - \mathbf{Z}_{n-1}^{(1)} | \mathcal{F}_{n-1}\big) = \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N}\left(\tfrac{N}{N}\right)^2 + \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(1)}}{N}\tfrac{N}{N} + \lambda\tfrac{N-\mathbf{Z}_{n-1}^{(1)}}{N}\left(\tfrac{\mathbf{Z}_{n-1}^{(1)}}{N}\right)^2 - \tfrac{\mathbf{Z}_{n-1}^{(1)}-\mathbf{Z}_{n-1}^{(2)}}{N},$$

for $1 < j < N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(j)} - \mathbf{Z}_{n-1}^{(j)} | \mathcal{F}_{n-1}\big) = \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N}\left(\tfrac{\mathbf{Z}_{n-1}^{(j-1)}}{N}\right)^2 + \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(j)}}{N}\tfrac{\mathbf{Z}_{n-1}^{(j-1)}}{N} + \lambda\tfrac{\mathbf{Z}_{n-1}^{(j-1)}-\mathbf{Z}_{n-1}^{(j)}}{N}\left(\tfrac{\mathbf{Z}_{n-1}^{(j)}}{N}\right)^2 - \tfrac{\mathbf{Z}_{n-1}^{(j)}-\mathbf{Z}_{n-1}^{(j+1)}}{N},$$

and for $j = N$:

$$\mathbb{E}\big(\mathbf{Z}_n^{(N)} - \mathbf{Z}_{n-1}^{(N)} | \mathcal{F}_{n-1}\big) = \lambda \frac{\mathbf{Z}_{n-1}^{(N-1)} - \mathbf{Z}_{n-1}^{(N)}}{N} \big(\frac{\mathbf{Z}_{n-1}^{(N-1)}}{N}\big)^2 + \lambda \frac{\mathbf{Z}_{n-1}^{(N-1)} - \mathbf{Z}_{n-1}^{(N)}}{N} \frac{\mathbf{Z}_{n-1}^{(N)}}{N} \frac{\mathbf{Z}_{n-1}^{(N-1)}}{N} + \lambda \frac{\mathbf{Z}_{n-1}^{(N)} - \mathbf{Z}_{n-1}^{(N)}}{N} \big(\frac{\mathbf{Z}_{n-1}^{(N)}}{N}\big)^2 - \frac{\mathbf{Z}_{n-1}^{(N)}}{N} .$$

Again, the associated ODE we get is (this time dropping the (t) s that we should really write for clarity!)

$$\frac{\mathrm{d}\mathbf{x}^{(1)}}{\mathrm{d}t} = \lambda\Big(\big(1 - \mathbf{x}^{(1)}\big) + \big(1 - \mathbf{x}^{(1)}\big)\mathbf{x}^{(1)} + \big(1 - \mathbf{x}^{(1)}\big)\big(\mathbf{x}^{(1)}\big)^2\Big) + \mathbf{x}^{(2)} - \mathbf{x}^{(1)} ,$$

for $1 < j < N$

$$\frac{\mathrm{d}\mathbf{x}^{(j)}}{\mathrm{d}t} = \lambda\Big(\big(\mathbf{x}^{(j-1)} - \mathbf{x}^{(j)}\big)\big(\mathbf{x}^{(j-1)}\big)^2 + \big(\mathbf{x}^{(j-1)} - \mathbf{x}^{(j)}\big)\mathbf{x}^{(j)}\mathbf{x}^{(j-1)} + \big(\mathbf{x}^{(j-1)} - \mathbf{x}^{(j)}\big)\big(\mathbf{x}^{(j)}\big)^2\Big) + \mathbf{x}^{(j+1)} - \mathbf{x}^{(j)} ,$$

and for $j = N$

$$\frac{\mathrm{d}\mathbf{x}^{(N)}}{\mathrm{d}t} = \lambda\Big(\big(\mathbf{x}^{(N-1)} - \mathbf{x}^{(N)}\big)\big(\mathbf{x}^{(N-1)}\big)^2 + \big(\mathbf{x}^{(N-1)} - \mathbf{x}^{(N)}\big)\mathbf{x}^{(N)}\mathbf{x}^{(N-1)} + \big(\mathbf{x}^{(N-1)} - \mathbf{x}^{(N)}\big)\big(\mathbf{x}^{(N)}\big)^2\Big) - \mathbf{x}^{(N)} .$$

As before, it's not too hard to prove that if we start this ODE with all coordinates positive, then they stay positive, and if they start decreasing in j, they stay that way. For the fixed point, with a bit of computer help we can come up with a guess. But it's helpful to notice that we can simplify the above equations a bit: what we see in the conditional expectation of queues increasing is a geometric series: the whole thing is a factor $\lambda\big(\mathbf{x}^{(j-1)} - \mathbf{x}^{(j)}\big)$ multiplied by $(\mathbf{x}^{(j-1)})^2 + \mathbf{x}^{(j)}\mathbf{x}^{(j-1)} + (\mathbf{x}^{(j)})^2$, and this product comes to $\lambda\big((\mathbf{x}^{(j-1)})^3 - (\mathbf{x}^{(j)})^3\big)$. We can hope that what we're supposed to do is ensure that the first term cancels with the $-\mathbf{x}^{(j)}$ and the second with the $\mathbf{x}^{(j+1)}$, because this is what happened in the 2 choice case (as there, this isn't going to quite work for $j = N$, but we hope it will provide an upper bound).

That means again we ought to guess that $\mathbf{x}^{(j)}$ is going to be a power of λ; we should take $\mathbf{x}^{(1)} = \lambda$, and then we'll want $\mathbf{x}^{(2)} = \lambda^4$, $\mathbf{x}^{(3)} = \lambda^{13}$, and so on: each time we triple and add one to the exponent. This is an easy recurrence to solve; the jth exponent should be $\frac{1}{2}\big(3^j - 1\big)$.

So our guess for the fixed point is it should be close to $\mathbf{x}^{(j)} = \lambda^{(3^j-1)/2}$. We can try to prove that if our ODE solution starts $0 \le \mathbf{x}^{(j)} \le \lambda^{(3^j-1)/2}$ for all j, then it will stay there: indeed, this works very much as for the 2 choice case.

So we end up aiming to solve a rather similar equation as for the two choice case:

$$\lambda^{(3^j-1)/2} = N^{-0.2}$$

and this solves as $j = \frac{1}{\log 3} \log\big(1 + 2\frac{0.2 \log N}{\log \lambda^{-1}}\big)$.

This looks quite a bit like the 2 choice solution. In some sense it genuinely is quite a bit better: we get to divide by $\log 3$ not $\log 2$. But it is still something which scales as $\log\log N$; we haven't made an improvement to the kind of function we get, just to the constant terms.

At this point, we can guess how the pattern will continue: the ODE will keep conveniently factorising, so that for s choices we'll have a fixed point guess which still looks like λ to some power. It turns out (it's not hard to check, either!) that the pattern is that the

exponent solves the multiply-by-s-and-add-one recursion, just as for 2 and 3 choices; this still gives us a geometric series to sum, which we can do: and the final result is that we get $j = \frac{1}{\log s} \log\left(1+(s-1)\frac{0.2\log N}{\log \lambda^{-1}}\right)$. Again, we keep getting slightly better solutions (the division by $\log s$ in front 'beats' the extra $s-1$ in the log), but not dramatically. 2 choices is good enough and (where it's public knowledge!) this is what's used in practice.

The way we analysed these processes—using differential equations—has good points and bad points. On the positive side, if you are Google this analysis is easy to modify and simulate to test scenarios: what happens if briefly the rate of jobs coming in goes up sharply? How quickly do we clear the queues if there is a quiet time? How fast will we recover from a systems problem which makes some or all servers fail to complete jobs for a while? And: if we do not like the answers to some of these questions, can we modify the system to improve, and how?

The negative side is that the Differential Equations Theorem has 'errors' that grow quite fast in time. For the infection process in the previous set of notes, this wasn't really an issue: we could analyse there (and here too!) the process until well after it gets very close to equilibrium, and in practice the interest in an infection process is to see how we can intervene before equilibrium is reached; there the realistic time scale is several months.

But for this load balancing process, in practice we get very close to equilibrium in a matter of seconds, and in fact our analysis, for realistic numbers and rates of jobs coming in, has the errors grow out of control in less than a minute. This doesn't mean the actual random processes go out of control, simply that our analysis stops being useful.

Furthermore, we couldn't really say anything about the single worst case queue. Google genuinely doesn't care all that much about this, because if customers extremely rarely have a bad experience, they're likely to believe it's the fault of their internet connection not Google. But it would be nice to know what we actually expect the worst case to be.

With all the processes we looked at, our analysis was not 'optimal'. One can usually improve, by taking in to account special features of a process, for example by using different concentration inequalities that perform better for the specific process: there are many different possibilities. If a process doesn't fit the form of the Differential Equations Theorem, nevertheless one often can deal with it. One common example is that for some processes, the conditional expectations aren't actually given by a well-behaved function of state and time, because they exhibit too much random fluctuation themselves: but one can sometimes prove that nevertheless averages of conditional expectations are well-behaved and this is enough to push an analysis through.

In short, if you feel like you want to use the Differential Equations Theorem, but it doesn't do what you need, rather than give up and look for something different, you should try to do the analysis a bit more by hand, using the same ideas.

Exercise 3. *Returning to the choice-of-2 Example 2, show that the fixed point of the ODE is stable. Using a catch-and-throw analysis, argue that in fact the load balancing process is likely to remain close to the fixed point for much longer than $N \log N$ steps.*

Hint: you will need to think quite hard about how to deal with the 'error' for long queues.

Solution to Exercise 3:

This exercise is quite hard (and I'm not giving the solution).

There are two main things you need to do. First, you need to prove some stability for the ODE. This is not all that easy, because it's in a high dimension.

Second, you need to improve the Differential Equations Theorem. In that, we used Azuma's inequality. This inequality, more or less, gives the 'worst possible' probabilistic estimate for a martingale sum. It turns out the 'worst possible' more or less corresponds to independent random variables taking values B and $-B$ with equal probability (B is as in the Differential Equations Theorem). But for the number of long queues, reality is quite far from this: we know that (assuming most queues are short) we're very unlikely to add to a long queue. There are other probabilistic inequalities that we can replace Azuma with, for example Freedman's inequality, which will let us aregue that the long queues cannot get nearly as big as the analysis above; and you need to know this to make the catch-and-throw method work (I think!).

There are other ways to approach this problem which will show that the long-term behaviour of the system is to converge to the stationary point of the ODE and (with a very good probability) will likely be close to it at any given time; these ways also let us properly show that the single worst queue is likely to be 'as you would expect' from the ODE solution, i.e. roughly $\log\log N$. However, especially given we already proved the Differential Equations Theorem, those ways require quite a lot more work, and also require more 'guessing parameters'—that is, making a correct choice of probability or expectation bounds which we can prove by induction, or complicated analysis of 'witness trees' which explain how an unlikely event can occur.

13 Markov chains

The infection process, and both processes above, have a special feature: the systems we are dealing with do not have a time dependence. The state at step n depends only on the state at step $n-1$, and not the value of n or the history of the process (the states at times $0, 1, \ldots, n-2$). This is what it means to say a process is *Markov*. (This isn't the universally accepted definition, though it certainly is the most common one; 'Markov' is sometimes used when there is a time dependence but not a history dependence, and also sometimes when there is some limited history dependence..! so be careful with the literature.)

We can define a Markov process with a general measurable *state space*, but rather often we're actually interested in a finite state space: here the theory is considerably simpler, and this is what corresponds to reality.

Definition 3 (Markov process, Markov kernel, Markov chain, transition matrix)**.** *Let (E, Σ) be a measurable space, and let $\kappa : \Sigma \times E \to [0, 1]$ be a function.*

Suppose that for each $S \in \Sigma$ the function $x \to \kappa(S, x)$ is Σ-measurable on E, and that for each fixed $x_0 \in E$ the function $S \to \kappa(S, x_0)$ defines a probability measure on (E, Σ). Then we say κ is a Markov kernel*.*

Suppose $\mu_0, \ldots$ is a sequence of probability measures on (E, Σ), and suppose that for each

$n \in \mathbb{N}$ and $A \in \Sigma$ we have

$$\mu_n(Y) = \int_Y \kappa(Y, x)\, \mathrm{d}\mu_{n-1}(x)\,.$$

Then we say $(\mu_n)_{n\geq 0}$ is a Markov process generated by κ, *with* initial distribution μ_0.

If E is finite, we can take $\Sigma = \mathcal{P}(E)$, and replace κ with a transition matrix A, *whose rows and columns are indexed by E, whose entries are in $[0,1]$, and which satisfies:*

For each $x_0 \in E$, the function $y \to A_{yx_0}$ defines a probability measure on (E, Σ); equivalently $\sum_{y\in E} A_{yx_0} = 1$.

Then we can write

$$\mu_n(y) = \sum_{j\in E} A_{yj}\mu_{n-1}(j)\,.$$

and $(\mu_n)_{n\geq 0}$ is a Markov chain.

Here we're using 'Markov process' for the general setup, and 'Markov chain' specifically for finite state spaces. This is not entirely standard; be careful when reading the literature.

Exercise 4. *Prove that if A is a transition matrix and μ is any probability measure on $(E, \mathcal{P}(E))$, then $\mu'(S) = \sum_{y\in S}\sum_{j\in E} A_{yj}\mu(j)$ is also a probability measure on $(E, \mathcal{P}(E))$.*

Solution to Exercise 4:

Observe that μ' is by definition finitely additive: since E is finite, this implies vacuously countable additivity.

It suffices to show that μ' takes non-negative values, that $\mu'(\emptyset) = 0$, and $\mu'(E) = 1$. The former two are trivial: since the entries of A are non-negative, the sum defining μ' is a sum of non-negative numbers so is non-negative; and for the empty set it is an empty sum so evaluates to 0. For the latter, we have

$$\mu'(E) = \sum_{y\in E}\sum_{j\in E} A_{yj}\mu(j) = \sum_{j\in E}\sum_{y\in E} A_{yj}\mu(j) = \sum_{j\in E}\mu(j)\sum_{y\in E} A_{yj} = \sum_{j\in E}\mu(j) = 1$$

as required.

It is normal, when dealing with Markov chains, to think of μ_n as a column vector of entries in $[0,1]$ whose sum is 1; this is the same thing as a probability measure on E. This gives us the nice linear algebra formula

$$\mu_n = A\mu_{n-1}$$

which defines the Markov chain. We'll rely on our knowledge of linear algebra to develop the theory: the critical point is that we obtain the simple formula $\mu_n = A^n\mu_0$ for the probability measure after n steps directly in terms of the initial probability measure.

However, we also often want a visual representation. The *transition digraph* of A is a directed graph (with loops) on E, putting an arc $\overrightarrow{ji}$ whenever $A_{ij} > 0$; we often say the *weight* of $\overrightarrow{ij}$ is A_{ji}.

We say that j is *reachable* from i if there is a directed path in the transition digraph from i to j. Equivalently, there is a sequence of positive probability transitions starting with i and

ending with j. If i is reachable from j and vice versa, we say i and j *communicate*; this is an equivalence relation. If E is the (unique) communicating class of A, we say the chain is *irreducible*.

A set S of states is called *absorbing* if there is no arc in the transition digraph leaving S. (Often we might talk about $a \in E$ being an *absorbing state* if $\{a\}$ is absorbing: that implies that the only arc $\overrightarrow{ax}$ in the transition digraph is the loop $\overrightarrow{aa}$ with weight 1.) An absorbing set S which is also a communicating class is a *terminal set*.

A *closed walk* of length k is a sequence of k not necessarily distinct vertices in the transition digraph, such that there is an arc from each vertex in the sequence to the next, and from the final vertex to the first. The *periodicity* of x is the greatest common divisor of the lengths of all closed walks containing x.

Theorem 4. *Let A be the transition matrix of a Markov chain. Then we have the following:*

All eigenvalues of A have absolute value at most 1, and 1 is an eigenvector of A. If A is symmetric, additionally all eigenvalues are real.

If s is not in any terminal set, then for all t we have $\lim_{n\to\infty} A^n_{st} = 0$.

The geometric multiplicity of 1 as an eigenvalue is equal to the number of terminal sets in A, and a non-zero eigenvector with eigenvalue 1 is strictly positive (or strictly negative) on at least one terminal set.

If x and y are in the same communicating class, they have the same periodicity.

For each $k \geq 2$, each terminal set of periodicity k contributes 1 to the geometric multiplicity of $e^{2\pi i\ell/k}$ for each $1 \leq \ell \leq k$; there are no other contributions.

The 'linear algebra' bits of this proof aren't examinable.

Proof. We use the fact that the eigenvalues of A and A^T are the same (because they have the same characteristic polynomial). Observe that since every column of A sums to 1, every column of $A - I$ sums to zero. Thus the sum of the rows of $A - I$ is the zero row, so $\det(A - I) = 0$ and 1 is an eigenvalue of A. Suppose now $\mathbf{e}$ is a non-zero eigenvector of A^T with eigenvalue $\lambda > 1$. Let j maximise $|\mathbf{e}_j|$, and without loss of generality (multiplying by a number of absolute value 1 if necessary) assume $\mathbf{e}_j$ is real and positive. Then we have

$$\lambda \mathbf{e}_j = (A^T\mathbf{e})_j = \sum_{i=1}^n A^T_{ji}\mathbf{e}_i = \sum_{i=1}^n A^{ij}\mathbf{e}_i \leq \sum_{i=1}^n A_{ij}|\mathbf{e}_i| \leq \sum_{i=1}^n A_{ij}\mathbf{e}_j = \mathbf{e}_j\,,$$

so $\lambda \leq 1$. Here we used the fact that the entries of A_{ij} are real and non-negative for the two inequalities.

If A is symmetric, it is a standard fact of linear algebra that all eigenvectors of A are real (and there is a basis of orthogonal eigenvectors).

For the limit statement, consider the set S of states which are not in any terminal set. For each $s \in S$, there is a path from s to a state in a terminal set, because the set of states reachable from s is an absorbing set (by definition) and contains a communicating class. So for each $s \in S$, fix such a path; let k be the maximum length of the fixed paths, and $\rho > 0$ the minimum probability of traversing any path (this probability is the product of

the transition probabilities on the path). Consider the random walk starting at $t \in E$ with transition matrix A. At each time $0, k, 2k, \ldots$, there are two possibilities. Either the random walk has reached a terminal set, or it has not. In the former case, by definition of k and ρ, the conditional probability of reaching a terminal set within the next k steps is at least ρ. Since the random walk stays in a terminal set once it enters it, the probability it is not in any terminal set at time ℓk is at most $(1-\rho)^\ell$. In particular, this is an upper bound on the probability that the random walk is at any given $s \in S$ at any time after ℓk. Since $(1-\rho)^\ell$ tends to zero as $n \to \infty$, we conclude the probability that the random walk is at s tends to zero as $n \to \infty$. But this probability is just A^n_{st}.

Let $C_1, \ldots, C_t$ be the terminal sets of A. For each i, looking at the submatrix A_i of A with rows and columns in C_i, we observe that all column sums of A_i equal 1 because these are the only non-zero entries in the column, by definition of 'absorbing'. Applying the previous part to each A_i, we obtain t eigenvectors $\mathbf{e}^{(1)}, \ldots, \mathbf{e}^{(t)}$ of A, each with eigenvalue 1, and whose supports are the pairwise disjoint sets $C_1, \ldots, C_t$. By the Perron-Frobenius Theorem, we may assume these eigenvectors have only non-negative entries. These eigenvectors are trivially pairwise orthogonal, so the multiplicity is at least t. Suppose now there is some eigenvector $\mathbf{e}$ with eigenvalue 1 which is not in the span of these t eigenvectors. We claim $\mathbf{e}$ is supported on the terminal sets: indeed, $\mathbf{e} = A^n\mathbf{e}$ for all n, and the rows of A which are not indexed by a terminal set state tend to zero as $n \to \infty$. Since $\mathbf{e}$ is not in the span of the given eigenvectors, there is some i such that the restriction of $\mathbf{e}$ to C_i is not a scalar multiple of $\mathbf{e}^{(i)}$. We claim the restriction $\mathbf{e}'$ of $\mathbf{e}$ to C_i is also an eigenvector of A with eigenvalue 1: this follows since there are no arcs in either direction between C_i and any other C_j, and $\mathbf{e}$ is zero on the vertices not in any C_j. So any linear combination of $\mathbf{e}'$ and $\mathbf{e}^{(i)}$ is an eigenvector of A with eigenvalue 1 supported on C_i. In particular, we can choose α such that $\mu = \mathbf{e}^{(i)} + \alpha\mathbf{e}'$ has no negative entries, at least one positive entry indexed by $x \in C_i$, and there is an entry indexed by $y \in C_i$ which is equal to zero. Rescaling if necessary, we can assume μ is a probability distribution. We have $A^n\mu = \mu$ for all n by definition, but at the same time y is reachable from x, so that there must exist some n such that $A^n\mu$ takes a positive value at y. This also proves that $\mathbf{e}^{(i)}$ itself is strictly positive on C_i.

For the periodicity, suppose x has period k. By definition, there is a path from x to y and from y to x; putting these paths together we have a closed walk C through x whose length is therefore divisible by k. Suppose now there is a closed walk C' through y whose length is not divisible by k: following C from x to y, then C', then C back from y to x we have a closed walk through x whose length is not divisible by k. This is a contradiction; we conclude the periodicity of y is divisible by k. A similar argument, using closed walks through x whose lengths have greatest common divisor k, gives that the periodicity of y is exactly k.

For the final part, as before we can argue that if $\mathbf{e}$ is an eigenvector with eigenvalue $e^{2\pi i\ell/k}$ of absolute value 1 then $\mathbf{e}$ is supported on the terminal sets; and then without loss of generality we can consider $\mathbf{e}$ restricted to a given C_i, which remains an eigenvector. Furthermore $\mathbf{e}$ is an eigenvector of A^k with eigenvalue 1. Each terminal set of periodicity p is split into p distinct terminal sets in A^p (corresponding to distance from a given vertex in the set modulo p, with all arcs going from one remainder class to the next). From this we obtain by the classification of eigenvalue 1 the upper bound on multiplicity; for the lower bound, if C_i has periodicity k and is split into sets $A_0, \ldots, A_{k-1}$ according to remainder, then

we can multiply the entries of $\mathbf{e}^{(i)}$ in A_j by $e^{j\cdot 2\pi i\ell/k}$ to obtain an eigenvector with eigenvalue $e^{2\pi i\ell/k}$; this constructs the required eigenvectors to match the upper bound. $\square$

While it's nice to know some of these facts, we will usually be interested in Markov chains which are irreducible. In this situation the eigenvalue 1 has geometric multiplicity 1. If in addition the Markov chain has periodicity 1 (since there is only one communicating class, this statement makes sense) then there are no other eigenvalues of absolute value 1.

A *stationary distribution* of A is a probability distribution μ such that $A\mu = \mu$. Every transition matrix A has at least one stationary distribution; it has exactly one stationary distribution if and only if there is exactly one terminal set, in which case the support of the stationary distribution is exactly the terminal set (that is, states not in the terminal set have probability zero in the stationary distribution, and states in the terminal set have positive probability).

If A is irreducible, the stationary distribution has positive probability on all states. However, this is not enough to guarantee that any Markov chain with transition matrix A will converge to the stationary distribution: for example $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is irreducible and the stationary distribution is $\binom{0.5}{0.5}$, but no other distribution converges (obviously!) to the stationary distribution.

Definition 5 (ergodic)**.** *If A is irreducible and has periodicity* 1*, we say A is* ergodic*.*

From what we now proved, it follows that an ergodic transition matrix has 1 as an eigenvalue with multiplicity 1, and all other eigenvalues have absolute value strictly smaller than 1.

Theorem 6. *Let A be an n by n ergodic matrix with stationary distribution π. Let λ be the largest absolute value of any eigenvalue other than* 1*. Then for any μ_0, letting $\mu_t = A^t\mu_0$ we have $\|\mu_t - \pi\|_\infty \le t^n\lambda^{t-n}$.*

This is a rather crude bound (which we're not going to prove!) but it does give us exponentially fast convergence to the stationary distribution.

However, this 'exponential' is much less practically useful than we might like. If the state space is large, or λ is very close to 1, or both, for this bound to make the distance usefully small requires taking t impractically large.

Exercise 5. *Determine the stationary distribution π for the infection process of Notes 4 Example 17. How does this compare to the stationary point of the associated ODE as ρ and λ vary?*

Suppose $\lambda = \frac{1}{2}$ and $\rho = \frac{1}{4}$, and $Z_0 = 0.75N$ (i.e. the initial distribution μ_0 has probability 1 *of being at $0.75N$, and* 0 *of any other value). Determine a lower bound on the smallest number of steps m such that*

$$\|\mu_m - \pi\|_\infty < 0.1\,.$$

By considering a sequence of steps leading to the stationary distribution, determine an upper bound on m. Hint: What is the simplest way this Markov chain might reach the stationary distribution? You should not expect to obtain an upper bound that is particularly close to your lower bound.

Solution to Exercise 5:

This one is easier than it looks. For every $1 \le n \le N$, there is an arc from n to $n-1$ and (if $n < N$) to $N+1$ in the transition digraph, so $\{1, \dots, N\}$ forms a communicating class. Since there is an arc from 1 to 0 in the transition digraph, and the only arc at 0 is a probability 1 loop, the unique terminal set is $\{0\}$ and the stationary distribution has probability 1 at of being at 0.

What you are supposed to notice is that this is *not* the ODE stationary point if $\lambda > \rho$, so that we can be sure that eventually this process will *not* track the ODE solution.

Consider the $\lambda = \frac{1}{2}$ and $\rho = \frac{1}{4}$ example, with $Z_0 = 0.75N$. This is what we analysed in Exercise 12 of the previous notes, and we argued that the process is likely to stay close to $0.5N$ (to be precise, in $[0.1N, 0.9N]$ with probability at least 0.99); but we could easily have made that be $[0.499N, 0.501N]$ or so) for time e^{cN} for some $c > 0$. So, rather surprisingly, what we see is that even though the process 'looks like' it reached some equilibrium state and stays there forever, in fact it doesn't. This e^{cN} (we copuld copy the c from the previous solutions, but let's not bother!) is our lower bound on m: if the process is in $[0.1N, 0.9N]$ with probability at least 0.99 at time $n = ce^N$, then in particular $\|\mu_n - \pi\|_\infty \ge 0.99$, because there can be probability at most 0.01 of being at 0, whereas the stationary distribution is to be at 0 with probability 1.

The easiest way we can reach the stationary distribution is simply to have a sequence of transitions 'downwards': no new infections occur and every step a person recovers. The probability of one such step is at least $(1-\lambda) \cdot \frac{1}{N}$ (if there is only one infected person, the probability of picking them to recover is $\frac{1}{N}$), and so even in the 'worst case' of starting at N infected people, the probability of reaching 0 in N steps is at least $\big((1-\lambda)/N\big)^N$.

As discussed in the solution to Exercise 17 (this wasn't asked in the exercise!) we can put an upper bound on the probability that $Z_{kN} \neq 0$ using this: the probability is at most

$$\Big(1 - \big((1-\lambda)/N\big)^N\Big)^k,$$

which we obtain as a product of conditional probabilities of not being at 0 at time jN given we are not at 0 at time $(j-1)N$, for $j = 0, \dots, k-1$. Here we're using that there is no 'other' way to not be at zero: $\{0\}$ is a terminal set of the Markov chain.

Using the same exponential approximation $1 - x \le e^{-x/2}$, valid for $0 \le x \le 1$, we get an upper bound

$$\mathbb{P}\big(Z_{kN} \neq 0\big) \le \exp\big(-\tfrac{1}{2}k(1-\lambda)^N N^{-N}\big)$$

and putting $k = 20N^N(1-\lambda)^{-N}$ this comes out equal to $e^{-10} < 0.1$. This gives the required upper bound on m, as $20N^{N+1}(1-\lambda)^{-N}$.

By being somewhat more careful, we could obtain a better upper bound e^{CN}, an exponential in N 'matching' our lower bound, as opposed to 'losing' a little bit and getting $e^{CN \log N}$, which is what the above bound is. In this context, even though e^{cN} and e^{CN} are really very far apart if $c < C$ and N is large, we'll usually think of those two functions as relatively 'good' bounds; they're at least both of the same form, as opposed to e^N versus e^{N^2} say. We usually can't hope to do anything very much better.

If you want to be 'more careful', the route is to notice that when Z_n is linear in N, then (as we argued in Notes 4) the probability of a downwards step is a constant bounded away from zero, not $1/N$ as we used above. When Z_n becomes small, the probability of a downwards step does genuinely get smaller, *but* λ is quite an overestimate of an upwards step (that's the bound we used in the above) because it's quite unlikely that our 'first person' (person a in the model) selected for the meeting will be infected. In fact, the probability of selecting an infected person a is the same as the probability of taking a downwards step. We can use that to argue that in any interval of time starting at n_0 of length roughly $N \log N$ (you need to be somewhat careful to choose the time correctly!), whatever the starting value $Z_{n_0} > 0$ is, it's not too unlikely that we select as person c (to recover!) in total at least N infected people, and as person a in the same time at most $2N$ infected people; and at each of the $2N$ time steps where we happened to select an infected person a, the $1-\lambda$ probability of non-infection is what happens.

More or less, our argument for an $e^{CN \log N}$ upper bound 'forces' a lot of recoveries very quickly, and this is 'too unlikely'. It's better to let $N \log N$ time steps pass not N, because this is the 'expected' amount of time for every individual in the population to be selected as person c; we're not 'forcing' anything; this is 'the most likely' route to reaching zero infected people. The cost of this, if we're not careful with the analysis, though, would be we need to ask for the $1-\lambda$ probability of non-infection to occur at all of those time steps. In the above, we're careful to explain why this isn't actually needed. (What we didn't do is fill in the details properly! To do so, the easiest route is to use Hoeffding's inequality to argue that the numbers selected are likely to be roughly accurate.)

Exercise 6. *Explain why the Markov chains of Examples 1 and 2, with $\lambda = 0.9$, are ergodic. They have different stationary distributions. For each example, suppose π is a stationary distribution: this is a distribution on $\{0, \ldots, N\}^N$. In both cases, try to give 'good' upper bounds on the stationary probabilities of long queues. You will need to guess at values which might work and use induction to argue that they do work.*

Solution to Exercise 6:

The first part is easy; though there are two 'different' answers. We need to decide what the state space of our Markov chain is; it can either be recording the current queue length for each server (which is a vector in $\{0, \ldots, N\}^N$) or it can be recording the number of servers with queue length at least J, for $1 \leq j \leq N$ (which is a vector in $\{0, \ldots, N\}^N$ whose components are non-increasing). Either way, from any given point in the state space, we can return to the all-zeroes vector with positive probability by a sequence of at most N^2 steps in which no customers arrive and a job is completed on every step, and we can from the all zeroes vector reach any given point in the state space by a sequence of at most N^2 steps in which no jobs are completed and customers happen to randomly pick (a) or pairs of the form (a, a) (for the two examples respectively) as required to obtain the desired assignment of jobs to servers. This verifies that the state space is a communicating class. The Markov chain has a self-loop at the all zeroes vertex (no customer arrives) and hence has periodicity 1; it is therefore ergodic.

For the second part, it's really better to think of the counting servers state space that we worked with (not least because we already thought about this more), and we'll stick to that.

A particularly nice class of Markov chains are *(lazy) graph random walks*. Let G be an (undirected) finite graph (multiple edges are allowed, but no loops). Starting at any given

vertex x (the vertices are the states of the Markov chain), we remain at x with probability ρ_x, and transition to each other vertex y such that xy is an edge of G with probability $\frac{1-\rho_x}{\deg(x)}$ (if $\deg(x) = 0$ then we require $\rho_x = 1$). The walk is called *lazy* if there exists $x \in G$ such that $\rho_x > 0$.

Exercise 7. *Prove that the lazy graph random walk, for a graph with at least two vertices, is ergodic if and only if $\rho_x < 1$ for all x and G is connected (there is a path between any two vertices), and either G contains an odd cycle or $\rho_x > 0$ for some $x \in G$.*

Let $\pi_x = \frac{1}{Z}\frac{\deg(x)}{1-\rho_x}$, where $\deg(x)$ is the number of edges (with multiplicity) at x and $Z = \sum_{x\in G}\frac{\deg(x)}{1-\rho_x}$. Prove that π is a stationary distribution of the lazy random walk.

Solution to Exercise 7:

For the first part, if G is not connected then there are two vertices x and y with no path between them; they are therefore in different communicating classes and the walk is not irreducible. If on the other hand G is connected, there is a path (automatically, in both directions) between any pair x and y, so there is only one communicating class and the walk is irreducible. So suppose now G is connected. If G has only one vertex x then $\rho_x = 1$ and the walk is trivially ergodic. If G has more than one vertex, letting x be a vertex there is an edge xy at x; now x, y, x is a closed walk and the periodicity of the Markov chain is at most 2. If $\rho_x > 0$ for some x or if there is an odd cycle in G, then this gives a closed walk of odd length, so that the periodicity of the Markov chain divides an odd number and can only be 1, i.e. the Markov chain is ergodic. If there is no odd cycle in G and $\rho_x = 0$ for all x, it is easy to see (this is a first result in graph theory) that the states partition into two sets and all transitions go between the two sets; so the Markov chain has periodicity 2 and is not ergodic.

For the second part, we simply do the matrix multiplication. Let A be the transition matrix of the lazy random walk.

We have

$$(A\pi)_x = \pi_x\rho_x + \sum_{y\in N(x)} \pi_y(1-\rho_y)\cdot\frac{1}{\deg(y)} = \frac{1}{Z}\frac{\rho_x\deg(x)}{1-\rho_x} + \sum_{y\in N(x)}\frac{1}{Z} = \frac{\deg(x)}{Z}\left(\frac{\rho_x}{1-\rho_x} + \frac{1-\rho_x}{1-\rho_x}\right) = \frac{\deg(x)}{Z(1-\rho_x)}$$

as required.

The 'point' of this is that we can easily see from 'local' information what the stationary distribution is, as opposed to the general case where the stationary distribution might depend a lot on states that are 'far away'. Furthermore, if we are designing the Markov chain we can easily adjust the desired stationary distribution simply by changing the laziness vector ρ.

The number Z is simply a normalising factor: Z is the traditional letter in this context, and in statistical mechanics it is called the *partition function* of the system.

14 Convergence of random variables

There are two types of 'convergence' of a sequence of random variables we can be interested in. One is 'convergence because it is a function'. The other is 'in distribution'.

We'll only look at one example of the latter type.

Definition 7 (Convergence in distribution)**.** *Let $X, X_1, \dots$ be a collection of random variables, defined on (possibly) different probability spaces. Recall the cumulative distribution function $F(x) = \mathbb{P}\big(X^{-1}((-\infty, x])$ and for each $n \in \mathbb{N}$ we have $F_n(x) = \mathbb{P}\big(X_n^{-1}((-\infty, x])$.*

We say $(X_n)_{n\in\mathbb{N}}$ converges in distribution *to X, written $X_n \xrightarrow{D} X$, if for every x such that $F(x)$ is continuous we have $\lim_{n\to\infty} F_n(x) = F(x)$.*

It's important here to remember that convergence in distribution is 'really' about random variables on completely different probability spaces; even if all the random variables happen to live on the same probability space, knowing something about $\big(X_n(\omega)\big)_{n\in\mathbb{N}}$ tells us nothing about $X(\omega)$. This is the 'convergence' of, for example, the Central Limit Theorem.

The remaining convergences that we will look at here (and we note that we barely scratch the surface: there are *many* different notions of convergence that are used; you pick whatever you can prove and work with..!) are all 'functional': all our random variables live on one probability space, and we expect to see that there is 'some kind of relation' between $X_n(\omega)$ and $X(\omega)$.

We've seen a couple of examples of 'function' convergence: *pointwise convergence* and *almost sure convergence*. We remind the definitions:

Definition 8 (pointwise, almost sure convergence)**.** *Let X and $(X_n)_{n\in\mathbb{N}}$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We say $(X_n)_{n\in\mathbb{N}}$* converges pointwise *to X if the sequence of numbers $\big(X_n(\omega)\big)_{n\in\mathbb{N}}$ converges to $X(\omega)$ for every $\omega \in \Omega$ (i.e. pointwise convergence,* surely*).*

We say the sequence converges almost surely *if $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ holds except possibly on a set of measure zero.*

Obviously pointwise (sure) convergence is a stronger property than almost sure convergence. Note that pointwise convergence doesn't actually require a probability space at all (it's nothing other than the pointwise convergence of functions one meets in a first or second Real Analysis course); while almost sure convergence does, *but* $\mathbb{P}$-almost sure and $\mathbb{Q}$-almost sure convergence are the same thing if $\mathbb{P}$ and $\mathbb{Q}$ have the same null sets (even if they are otherwise very different measures). We saw these notions around the Monotone Convergence Theorem and Dominated Convergence Theorem, and their versions for the conditional expectation; and the Radon-Nikodym Theorem told us when two σ-finite measures have the same null sets.

If we want to know about convergence of functions of random variables, almost sure convergence is usually what we want.

Exercise 8. *Suppose $f : \mathbb{R} \to \mathbb{R}$ is any continuous function, and $X_n \to X$ almost surely. Prove that $f(X_n) \to f(X)$ almost surely.*

Solution to Exercise 8:

Suppose that we have almost sure convergence, i.e. for $\omega \notin A$ we have $X_n(\omega) \to X(\omega)$, where $\mathbb{P}(A) = 0$. Given $\omega \notin A$, it follows (by the first course in analysis sequential characterisation of continuity) from continuity of f at $X(\omega)$ that $\lim_{n\to\infty} f(X_n(\omega)) = f(X(\omega)$ as required.

These two convergence notions are often not strong enough on their own to be very useful for integrals, though: see the 'standard counterexample' mentioned before Notes 2 Exercise 9. Indeed, in the two mentioned theorems, we used the notion with an extra property, either 'monotone increasing' or 'dominated'. However, often we are able to obtain one of these extra properties, and then we can 'upgrade' the less useful convergence to the more useful one.

Nevertheless, sometimes we can prove less (or, we can more easily prove less followed by using a standard theorem to upgrade it to a better notion of convergence).

Definition 9 (Convergence in probability)**.** *Let X and $(X_n)_{n\in\mathbb{N}}$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We say $(X_n)_{n\in\mathbb{N}}$* converges in probability *to X, and write $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$ we have $\lim_{n\to\infty} \mathbb{P}\big(|X_n - X| < \varepsilon\big) = 1$.*

It's a standard fact that almost sure convergence implies convergence in probability.

Exercise 9. *Suppose $(X_n)_{n\in\mathbb{N}}$ converges almost surely to X. By considering, for a given $\varepsilon > 0$, the sets (as n varies) $\bigcup_{m\geq n} \{\omega \in \Omega : |X_m(\omega) - X(\omega)| \geq \varepsilon\}$, argue that $X_n \xrightarrow{\mathbb{P}} X$.*

Solution to Exercise 9:

Given $\varepsilon > 0$, let $A_n = \bigcup_{m\geq n}\{\omega \in \Omega : |X_m(\omega) - X(\omega)| \geq \varepsilon\}$. Observe that if ω is such that $X_\ell(\omega) \to X(\omega)$ as $\ell \to \infty$, then by definition there exists N such that $\omega \notin A_N$. Thus $\bigcap_{n\in\mathbb{N}} A_n$ contains only ω such that $X_\ell(\omega) \not\to X(\omega)$, and by definition of almost sure convergence we see that $\bigcap_{n\in\mathbb{N}} A_n$ has measure zero.

By continuity of measure, it follows $\lim_{n\to\infty} \mathbb{P}(A_n) = 0$. Since this holds for all $\varepsilon > 0$, we have verified convergence in probability.

It's also standard that the converse implication fails.

Exercise 10. *Let $f_{k,n}(x) : [0,1] \to \mathbb{R}$, for each $n \in \mathbb{N}$ and $0 \leq k \leq 2^n - 1$, be defined by $f_{k,n}(x) = \mathbf{1}_{2^{-n}k \leq x \leq 2^{-n}(k+1)}$. Considering $[0,1]$ with its Lebesgue measure as a probability space, show that any sequence $\big(f_{k_i,n_i}\big)_{i\in\mathbb{N}}$ with $n_i \to \infty$ as $i \to \infty$ converges in probability to the constant zero function.*

Give an example of how to choose $(k_i, n_i)_{i\in\mathbb{N}}$ such that for every $x \in [0,1]$ we have $f_{k_i,n_i}(x) = 1$ infinitely often as $i \to \infty$, and explain why this implies that the sequence of random variables does not converge almost surely to any limit.

Solution to Exercise 10:

If $\varepsilon > 1$, then $\mathbb{P}(|f_{k,n} - 0| > \varepsilon) = 0$, while if $0 < \varepsilon \leq 1$ then $\mathbb{P}(|f_{k,n} - 0| > \varepsilon) = 2^{-n}$. Since $2^{-n} \to 0$ as $n \to \infty$, it follows that any sequence $\big(f_{k_i,n_i}\big)_{i\in\mathbb{N}}$ converges in probability to the constant zero function provided $n_i \to \infty$ as $i \to \infty$.

Consider the sequence

$$(0,1), (1,1), (0,2), (1,2), (2,2), (3,2), \dots ,$$

where for each $n \in \mathbb{N}$ in succession we let k run through the 2^n possible values from 0 to $2^n - 1$. We claim the corresponding sequence of functions, which does converge in probability

to the constant zero function since $n_i \to \infty$ as $i \to \infty$, does not converge almost surely to any limit.

Indeed, for any given $x \in [0,1]$, for any fixed n the value of $f_{k,n}(x)$ is 1 for at most 2 values of k, and 0 for all the rest. Thus $\big(f_{k_i,n_i}(x)\big)_{i\in\mathbb{N}}$ is a sequence whose terms are all in $\{0,1\}$, and both values occur infinitely often. Such a sequence is not convergent (for example, because it has a subsequence converging to 0 and another converging to 1): so not only does $\big(f_{k_i,n_i}\big)_{i\in\mathbb{N}}$ not converge almost surely to any limit, it actually does not converge pointwise at any $x \in [0,1]$ at all; the measure of non-converging x is 1.

While it doesn't make sense to ask if convergence in distribution implies one of the other forms of convergence (there is no reason why there should be a relation between $X_n(\omega)$ and $X(\omega)$, even if all the random variables are defined on $(\Omega, \mathcal{F}, \mathbb{P})$), we can ask the reverse question.

Exercise 11. *Use the union bound to prove that* $\mathbb{P}(X \le a) \le \mathbb{P}(Y \le a+\varepsilon) + \mathbb{P}(|X-Y| > \varepsilon)$ *for any random variables* X, Y.

Let X *and* $(X_n)_{n\in\mathbb{N}}$ *be random variables on* $(\Omega, \mathcal{F}, \mathbb{P})$*; suppose that* $X_n \xrightarrow{\mathbb{P}} X$. *Prove that* $X_n \xrightarrow{D} X$.

Solution to Exercise 11:

Suppose $\omega \in \Omega$ is such that $X(\omega) \le a$. We claim that either $Y(\omega) \le a+\varepsilon$, or $|X(\omega) - Y(\omega)| > \varepsilon$. Indeed, suppose the former is false: then $Y(\omega) > a+\varepsilon$, and subtracting $Y(\omega) - X(\omega) > \varepsilon$, so the second condition is true. This verifies that $\{X \le a\}$ is contained in $\{Y \le a\} \cup \{|X-Y| > \varepsilon\}$, and the desired probability claim follows by the union bound.

Given now $\varepsilon > 0$ and $a \in \mathbb{R}$, and any $n \in \mathbb{N}$, we have

$$\mathbb{P}(X \le a-\varepsilon) \le \mathbb{P}(X_n \le a) + \mathbb{P}(|X - X_n| > \varepsilon) \le \mathbb{P}(X \le a+\varepsilon) + 2\mathbb{P}(|X - X_n| > \varepsilon)$$

by applying the first part twice, once with X, X_n and $a-\varepsilon$, and once with X_n, X and a.

By definition of convergence in probability we have $\lim_{n\to\infty} \mathbb{P}(|X - X_n| > \varepsilon) = 0$; in particular, there is N such that for all $n \ge N$ we have $\mathbb{P}(|X - X_n| > \varepsilon) < \varepsilon$ Thus, rewriting this in terms of the distribution functions F of X, and F_n of X_n, we obtain that for all $\varepsilon > 0$ there exists N such that for all $n \ge N$ we have

$$F(a-\varepsilon) \le F_n(a) + \varepsilon \le F(a+\varepsilon) + 2\varepsilon\,.$$

Suppose now F is continuous at a: then as $\varepsilon \to 0$ the outside terms both tend to $F(a)$ and we obtain

$$F(a) = \lim_{n\to\infty} F_n(a)$$

verifying convergence in distribution.

The final collection of 'convergence notions' we'll look at are identical to some standard notions from analysis: convergence in L^p. In this $1 \le p \le \infty$ is a parameter; we consider a vector space of integrable functions on Ω, with the norm

$$\|f\|_p := \Big(\int_\Omega |f|^p \,\mathrm{d}\mathbb{P}\Big)^{1/p} = \Big(\mathbb{E}\big(|f|^p\big)\Big)^{1/p}$$

for $p < \infty$, while $\|f\|_\infty = \operatorname{ess\,sup}_{\omega\in\Omega} |f(\omega)|$. Here ess sup is the *essential supremum*,

$$\operatorname*{ess\,sup}_{\omega\in\Omega} |f(\omega)| = \inf\left\{c \in \mathbb{R} \,:\, |f(\omega)| \le c \text{ almost surely}\right\}.$$

and $L_p(\Omega, \mathcal{F}, \mathbb{P})$ is the collection of measurable functions f from Ω to $\mathbb{R}$ such that $\|f\|_p < \infty$. The point of the 'essential supremum' is that (as with the other norms) it gives the same value for any functions f and g which are almost surely equal.

It is not immediately obvious (but also not too hard to prove) that these functions indeed define 'norms up to almost sure equality'; the statement that the Triangle Inequality holds (which is the only non-trivial part) is Minkowski's Inequality: $\|f+g\|_p \le \|f\|_p + \|g\|_p$ for all $1 \le p \le \infty$ and $f, g \in L_p$. Formally they are *seminorms* (meaning we relax the requirement that $\|f\|_p = 0$ if and only if $f = 0$). The alternative here is to consider two elements of L_p to be equal if they are almost surely equal. This is the choice mentioned in Notes 2 we are *not* making in this course: here it would be somewhat cleaner, but we will here simply refer to 'the norm' even if this isn't formally quite accurate. We have $\|f\|_p = 0$ if and only if $f = 0$ holds almost surely, for all $1 \le p \le \infty$.

Definition 10 (Convergence in L_p)**.** *Let $1 \le p \le \infty$, and let X and $(X_n)_{n\in\mathbb{N}}$ be random variables in $L_p(\Omega, \mathcal{F}, \mathbb{P})$. We say $(X_n)_{n\in\mathbb{N}}$* converges in L_p *to X if $\lim_{n\to\infty} \|X_n - |\|_p = 0$.*

Theorem 11. *Let $1 \le p < q \le \infty$, and let X and $(X_n)_{n\in\mathbb{N}}$ be random variables in $L_q(\Omega, \mathcal{F}, \mathbb{P})$. Suppose (X_n) converges to X in L_q. Then also (X_n) converges to X in L_p.*

Proof. We need to do $q = \infty$ as a special case. First, if $\|X\|_\infty < \infty$, by definition there is C such that $|X| \le C$ holds almost surely, from which by monotonicity of the integral we have $\mathbb{E}(|X|^p) \le \mathbb{E}(C^p) = C^p\mathbb{P}(\Omega)$, so X (and similarly the X_n) are in L_p. For the convergence, we argue similarly: by definition, given $\varepsilon > 0$ there exists N such that for $n > N$ we have $\|X - X_n\|_\infty < \varepsilon$ almost surely, and by monotonicity we have $\mathbb{E}(|X - X_n|^p) \le \mathbb{E}(\varepsilon^p) = \varepsilon^p\mathbb{P}(\Omega)$, so $\|X - X_n\|_p \le \varepsilon$.

Now suppose $q < \infty$. It will be convenient to work with $\|\cdot\|_q^q$, and similarly the pth power of $\|\cdot\|_p$; these tend to zero if and only if the corresponding norms do. We deal with the convergence; the proof of being in L_p is basically the same. Given $c > 0$, we have

$$\begin{aligned}
\mathbb{E}\big(|X - X_n|^p\big) &= \mathbb{E}\big(|X - X_n|^p \mathbf{1}_{|X-X_n|\ge c}\big) + \mathbb{E}\big(|X - X_n|^p \mathbf{1}_{|X-X_n|<c}\big) \\
&\le \mathbb{E}\big(c^{p-q}|X - X_n|^q \mathbf{1}_{|X-X_n|\ge c}\big) + \mathbb{E}\big(c^p \mathbf{1}_{|X-X_n|<c}\big) \\
&\le c^{p-q}\mathbb{E}\big(|X - X_n|^q\big) + c^p\mathbb{P}(\Omega)\,,
\end{aligned}$$

where both inequalities use monotonicity of the integral. This inequality directly tells us that if $X - X_n$ is in L_q, then it is also in L_p. When n is sufficiently large (for any given $c > 0$) we have $\mathbb{E}(|X - X_n|^q) \le c^q$, so for any $c > 0$ and sufficiently large n we obtain $\mathbb{E}(|X - X_n|^p) \le 2c^p$. As $c \to 0$ we obtain the desired convergence. □

It's worth recalling that in the 'usual' function setting (on all of $\mathbb{R}$, or in general measure spaces) this theorem is false: in both parts, we used $\mathbb{P}(\Omega) = 1$ to obtain some inequalities, and (on the real line, for example) none of these versions of convergence implies another (though it *is* true that if we have containment and / or convergence in L_p and $L_{p'}$, then we have it in L_q too whenever $1 \le p < q < p' \le \infty$). In the proofs above, all our final upper

bounds on $\|\cdot\|_p$ have a term containing $\mathbb{P}(\Omega)$, and we rely on this being a finite number (in fact, it's 1, of course) for the containment in L_p and for the convergence; if we work on a non-finite measure space, this of course fails.

Exercise 12. *Fix $1 \le p < q \le \infty$, and let $(\Omega, \mathcal{F}, \mathbb{P})$ be Lebesgue measure on $[0,1)$. By considering $f(x) = (1-x)^\alpha$ and $f_n = n^\alpha \mathbf{1}_{[0,1/n)}$ for suitably chosen α, argue that there are functions in L_p which are not in L_q, and demonstrate that even if an L_p-convergent sequence of functions, and their L_p-limit, are in L_q, nevertheless it can be that the sequence does not converge in L_q.*

Solution to Exercise 12:

For the failure of containment, we use $f(x) = (1-x)^{-\alpha}$ with α chosen such that $\alpha p < 1 < \alpha q$. We have

$$\mathbb{E}\big(f^p\big) = \int_{x=0}^{1} (1-x)^{-\alpha p}\,\mathrm{d}x = \tfrac{-1}{1-\alpha p}\big[(1-x)^{1-\alpha p}\big]_{x=0}^{1} = \tfrac{1}{1-\alpha p}$$

so that $\|f\|_p < \infty$, but for any $0 < c < 1$ we have

$$\int_{x=0}^{c} (1-x)^{-\alpha q}\,\mathrm{d}x = \tfrac{-1}{1-\alpha q}\big[(1-x)^{1-\alpha q}\big]_{x=0}^{c} = \tfrac{1}{\alpha q-1}\big((1-c)^{1-\alpha q} - 1\big)$$

which diverges as $c \to 1^-$ since $1 - \alpha q < 1$.

For the failure of convergence, we claim that (for the same α) the functions f_n are in L_p and L_q, their L_p-limit is the constant zero function (which is also in L_p and L_q), but the sequence does not converge in L_q. The containment is trivial: for any real-valued simple function g, $\|g\|_p$ is a finite sum of real numbers and so is $< \infty$. For the convergence in L_p, observe $\|f_n\|_p^p = n^{\alpha p - 1}$ which tends to zero as $n \to \infty$.

It's easy (and maybe in practice enough!) to prove that the sequence does not converge in L_q to the constant zero function: we have $\|f_n\|_q^q = n^{\alpha q - 1}$ which tends to ∞. However the exercise asks us to show there is no L_q-limit at all. So suppose $g \in L_q(\Omega, \mathcal{F}, \mathbb{P})$ is any function, and consider

$$\|f_n - g\|_q \ge \|f_n\|_q - \|g\|_q = \|f_n\|_q - \|g\|_q$$

which tends to ∞ as $n \to \infty$, since $\|g\|_q$ is a constant. Here the inequality is the 'reverse triangle inequality' obtained by applying the triangle inequality to $(f_n - g) + g$.

The 'point' here is that the L_q-norm 'blows up' very large values that occur infrequently more than the L_p-norm does. On the real line, conversely the L_q-norm flattens the 'tails' of very small values that occur on a large part of the space more than the L_p-norm does; it's not too hard to develop this intuition into a proof that on the real line, L_q-containment or convergence does not in general imply L_p-containment or convergence. Again, what makes the probability space special is that the measure of the space is bounded: small values on a set A can't 'blow up' the norm or spoil convergence by being multiplied by an extremely large $\mu(A)$, because $\mu(A)$ is bounded by $\mu(\Omega) = 1$. In general convergence in $L_p(\Omega, \mathcal{F}, \mathbb{P})$, for any $p < \infty$, does not imply almost sure convergence, or vice versa.

Exercise 13. *Using the example from Exercise 10, prove that convergence in L_p for any $1 \le p < \infty$ does not imply convergence almost surely.*

Prove that convergence in $\|\cdot\|_\infty$ does imply almost sure convergence.

Solution to Exercise 13:

For any $1 \leq p < \infty$, we have $\|f_{k,n}\|_p^p = 2^{-n}$, so any sequence f_{k_i,n_i} with $n_i \to \infty$ as $i \to \infty$ converges in L_p to the constant zero function; but in Exercise 13 we argued that there exists such a sequence which does not almost surely converge at all, and in particular not to the constant zero function.

For $p = \infty$, suppose that $(f_n)_{n\in\mathbb{N}}$ converges in L_∞ to f. Given any $\varepsilon > 0$, by definition there exists N such that for all $\omega \notin A_\varepsilon$, where $\mathbb{P}(A_\varepsilon) = 0$, we have $|f_n(\omega) - f(\omega)| < \varepsilon$. In particular, if $\omega \notin \bigcup_{m\in\mathbb{N}} A_{1/m}$, then $f_n(\omega) \to f(\omega)$ as $n \to \infty$. Since $\bigcup_{m\in\mathbb{N}} A_{1/m}$ is a countable union of measure zero sets, it has measure zero: this verifies almost sure convergence.

The 'standard counterexample' in Notes 2 (sometimes called the 'sliding bump') is an example of an almost surely converging sequence of functions which fails to converge in L_1, and so also L_p for any $p > 1$; the example from Exercise 12 also works.

The 'point' of convergence statements is to let us go back and forth between continuous and discrete random variables, and to give us a way to estimate quantities.

It is generally easier to prove 'analysis' statements about discrete random variables—especially simple functions!—and then use a Convergence Theorem to deduce a statement about general random variables. This was our standard strategy for proving most of our statements about integrals in previous parts! Similarly, if we want to use a computer to calculate numerically (perhaps with some kind of guarantee about errors!), even if we start with general functions, we are forced to calculate something to do with discrete functions: a convergence statement is needed to argue that what we calculate has something to do with the calculation we wanted to approximate. We won't go in to this much in this course (but see the next section).

On the other hand, if we want to calculate symbolically (integrate, for example!), it is usually far easier to work in the continuous setting: 'small errors' that we have to handle exactly or argue are really small in the discrete setting usually vanish in a 'continuous limit', and we have the standard calculus tools available. Again, in general we need some kind of convergence statement to argue that what we calculate in the continuous setting has something to do with the large N limit of the discrete setting that we wanted. We again won't go in to much detail here: but we should notice that we took full advantage of this in proving the binary Johnson-Lindenstrauss Theorem, doing the difficult moment generating function calculation in the 'continuous limit' Gaussian case, and using that to deduce the statement we really wanted for the discrete case. There, we avoided needing a convergence statement; we argued that the discrete case is strictly better than the Gaussian case (in a sense, we were lucky that this happened to be true). But we would still have been able to make the argument work if it hadn't been true: we would have argued that the discrete case converges to the Gaussian (and justified this properly! which requires some care, since it's only true when $\|\mathbf{x}\|_\infty$ tends to zero in the binary Projection Theorem!) and used some theorem to argue that this means the moment generating functions also converge.

The most important 'convergence theorem' for purposes of estimating quantities is Hölder's inequality:

Theorem 12 (Hölder's inequality)**.** *Let f, g be functions on $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose $1 \leq p, q \leq \infty$*

satisfy $\frac{1}{p} + \frac{1}{q} = 1$. *Then we have*

$$\|fg\|_1 \leq \|f\|_p \|g\|_q \,.$$

We won't prove this (it is a standard fact of analysis). The numbers p and q are sometimes called *Hölder* conjugates. It's also worth noting that if $1 < p, q < \infty$, then there is a characterisation of equality: if f or g is almost surely zero, we have equality; if not, we have equality if and only if $\alpha|f|^p = \beta|g|^q$ almost surely, where $\alpha, \beta > 0$ are constants. In particular, the case $p = q = 2$ is the *Cauchy-Schwarz inequality.*

To see why this is useful, suppose we know $(X_n)_{n\in\mathbb{N}}$ converges in L_p to X, and suppose Y is any random variable in L_q. Then we have

$$\|(X_n - X)Y\|_1 \leq \|X_n - X\|_p \|Y\|_q$$

and the right hand side converges to zero: so the left hand side does too. We might want to think of X_nY as a 'discrete' random variable and we are interested in, or can calculate it for, large n; while XY is the 'continuous limit'. What we now argued is that their expectations converge; if we can calculate one, we get the other too.

15 Monte Carlo estimation

This is going to be a short section; more or less, having set up the theory, we might as well mention what it can be used for—the topic is a huge one, and we will barely scratch the surface.

A fundamental problem in lots of areas of maths, and even more in applications, is to calculate integrals $\int_\Omega f \,\mathrm{d}\mathbb{P}$. Sometimes we can do this analytically, but usually we cannot; we are forced to do some kind of numerical approximations.

If f is a well-behaved function and Ω is a nice set—for example f is monotone increasing and $\Omega = [0, 1]$ with Lebesgue measure—then this is fairly easy and even comes with guarantees of accuracy. In this case, we would write

$$\sum_{k=0}^{N-1} \tfrac{f(k/N)}{N} \leq \int_\Omega f \,\mathrm{d}\mathbb{P} \leq \sum_{k=1}^{N} \tfrac{f(k/N)}{N} \,,$$

and we would see what N gives us a good enough answer.

If f isn't well-behaved enough, there is not much we can do 'generically'; if f can take very large values sporadically, we would need to do some analytic work to understand how and where, otherwise we simply won't get a good estimate. But in practice we tend to be interested in quite well-behaved functions.

However, if Ω is not a nice set, we still have trouble. For example, suppose $\Omega = [0, 1]^d$ with Lebesgue measure, where d is a very large number (as in the big data setting). Suppose f is monotone increasing in each coordinate. We still have, in principle, a nice way to estimate the integral, even with guarantees of accuracy:

$$\sum_{k_1,\ldots,k_d=0}^{N-1} \tfrac{f(k_1/N,\ldots,k_d/N)}{N^d} \leq \int_\Omega f \,\mathrm{d}\mathbb{P} \leq \sum_{k_1,\ldots,k_d=1}^{N} \tfrac{f(k_1/N,\ldots,k_d/N)}{N^d} \,,$$

In practice, though, this is useless; even if $N = 2$ would in principle be enough (it almost never is!) for the accuracy we need, we're looking at summing up 2^d numbers, and for $d = 1000$ (which is not particularly big data!) the calculation will be interrupted by the Sun's expansion to a red giant, or the electricity company shutting off the meter.

Theorem 13. *Suppose f is any measurable function on $(\Omega, \mathcal{F}, \mathbb{P})$ bounded by B. Let $\omega_1, \dots, \omega n$ be independent samples from Ω according to $\mathbb{P}$. Then with probability at least $1 - 2e^{-t^2 n/(2B^2)}$, we have*

$$\left| \tfrac{f(\omega_1)+\cdots+f(\omega_n)}{n} - \int_\Omega f \, \mathrm{d}\mathbb{P} \right| \leq t \, .$$

The mean of the sampled points $\frac{1}{n}\sum_{i=1}^n f(\omega_i)$ is a *Monte Carlo estimate of the integral.*

Proof. Observe that the $f(\omega_i)$ are independent identically distributed random variables (formally, on Ω^n with product measure), whose expectations are by definition $\int_\Omega f \, \mathrm{d}\mathbb{P}\mathbb{E}(f)$. Then Hoeffding's inequality states that for any $t > 0$ we have

$$\mathbb{P}\Big(\Big| \sum_{i=1}^n f(\omega_i) - n\mathbb{E}(f) \Big| \geq tn \Big) \leq 2 \exp\left(\tfrac{-2(tn)^2}{n(2B)^2} \right) . \qquad \square$$

An important point to note here is that we don't have any 'guarantees': no matter how big n is, we could always have chosen unluckily and obtained a bad estimate. However in practice we rapidly stop worrying: for quite small n, the chance of being unlucky in choice of samples is less than the chance of a cosmic ray striking the processor and messing up the calculation. The huge benefit we have is that we didn't need to know much about f (though, being bounded, especially with a 'reasonable' value of B, is quite an assumption!) and we didn't need to know anything about Ω apart from being able to sample from it according to $\mathbb{P}$. Returning to $\Omega = [0,1]^d$ with Lebesgue measure, if we can sample uniformly from a high-dimensional cube, we expect to get a very good estimate of $\int_\Omega f \, \mathrm{d}\mathbb{P}$ with 'few' samples, in particular there isn't any dependence on the dimension.

There is a lot of literature on this: what if we don't have boundedness? or if we do, but B is 'unreasonable' due to very rare large values, can we nevertheless prove something? The answer is often Yes, often simply by using a 'more careful' version of Hoeffding's inequality, such as one of the Bernstein inequalities.

In practice, being able to sample uniformly from a continuous distribution *is* an unreasonable assumption. A computer samples binary numbers; if f is supported only on irrationals, the computer's Monte Carlo estimation of $\int_\Omega f \, \mathrm{d}\mathbb{P}$ is guaranteed to be zero, but the true answer can be very different.

However, we aren't interested in functions like this in practice.

Exercise 14. *Suppose f is L-Lipschitz continuous and bounded by B on $[0,1]^d$. Suppose a computer has access to independent uniform random samples from the lattice $\{0, \frac{1}{N}, \dots, \frac{N-1}{N}\}^d$. Explain why with probability at least $1 - 2e^{-t^2 n/(2B^2)}$, we can obtain a Monte Carlo estimate of the integral with error at most $t + L/N$.*

Suppose that $N = 2^s$ for some $s \in \mathbb{N}$. Argue that it is enough for the computer to have access to independent uniform random bits (samples from $\{0,1\}$).

Solution to Exercise 14:

Observe that (ignoring a set of measure zero on the boundary!) $[0,1]^d$ is split up into N^d equal disjoint *lattice cubes*, translates of $[0,1/N)^d$ by translating the 0 point of the cube to the lattice points. Consider the experiment of selecting a uniform random point $\mathbf{x}$ in $[0,1]^d$, identifying the lattice cube containing it, and returning the corresponding lattice point $\mathcal{L}(\mathbf{x})$. We claim that this experiment will with probability 1 return a uniformly distributed lattice point. (With probability zero we picked a boundary point which isn't in any lattice cube.)

By Theorem 13, with probability at least $1-2e^{-t^2n/(2B^2)}$, when we sample n independent uniform points in $[0,1]^d$ we obtain

$$\left|\tfrac{f(\mathbf{x}_1)+\cdots+f(\mathbf{x}_n)}{n} - \int_\Omega f \,\mathrm{d}\mathbb{P}\right| \le t\,.$$

By L-Lipschitz continuity of f, we have $\left|f\big(\mathcal{L}(\mathbf{x})\big) - f(\mathbf{x})\right| \le L/N$: the lattice point $\mathcal{L}(\mathbf{x})$ differs from $\mathbf{x}$ by at most $\frac{1}{N}$ in each coordinate. It follows that

$$\left|\tfrac{f(\mathcal{L}(\mathbf{x}_1))+\cdots+f(\mathcal{L}(\mathbf{x}_n))}{n} - \int_\Omega f \,\mathrm{d}\mathbb{P}\right| \le t + L/N\,,$$

and the right hand side is the same experiment as picking n uniform random lattice points (which we assumed is possible) and returning their sample mean.

If we now only assume access to independent random bits, but $N = 2^s$ is a power of 2, then we can generate a number in $\{0,\dots,N-1\}$ by simply sampling a sequence of s random bits and interpreting the sequence as a binary number; so with ds random bits we obtain d independent random samples from $\{0,\dots,N-1\}$ which gives the uniform sample from the lattice points we need.

Note that in practice we need N to be quite large in order for the *discretisation error* L/N to be small, but this is 'cheap': N grows exponentially in the number of bits we use (which is the same order of magnitude as the calculation time we need).

It's also worth pointing out that we *cannot* generate genuinely uniform samples even from as simple a distribution as uniform on $\{1,2,3\}$ if we only have access to uniform random bits; even for very simple practical algorithms, we'll need to handle approximations if we want to implement them computationally.

If we are dealing with a more complicated distribution $\mathbb{P}$, we usually won't see an 'obvious' way to sample uniformly from it, or from some good enough discretisation (which is what Exercise 14 is doing).

For lack of time, we won't prove anything in this course, but we observe that we have seen two ways to deal with this situation.

If we are dealing with $\mathbb{P}$ being absolutely continuous with respect to some simpler distribution $\mathbb{Q}$ (like Lebesgue measure on $[0,1]^d$), and we can compute the Radon-Nikodym derivative $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}$, then we can use the change of measure formula to write

$$\int_\Omega f \,\mathrm{d}\mathbb{P} = \int_\Omega f \tfrac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} \,\mathrm{d}\mathbb{Q}$$

and do Monte Carlo estimation of the latter integral.

Often however we don't have such a $\mathbb{Q}$ to hand, or we cannot compute the Radon-Nikodym derivative. If we are working on a discrete probability space (perhaps a good discretisation of a continuous space) an alternative is to construct an ergodic Markov chain whose stationary distribution is $\mathbb{P}$. We then get *Markov chain Monte Carlo*.

In theory, this is simple: run the Markov chain until we are guaranteed 'close enough' to $\mathbb{P}$, treat the output element of Ω as our first sample from $\mathbb{P}$, repeat n times and argue that our errors are likely to be small.

In practice, there are pitfalls. We need to design a Markov chain which we *can* easily run—ideally, that means transition probabilities are powers of 2 and easy to compute.

We need the Markov chain to be *rapidly mixing*—that is, it is guaranteed to converge 'fast'.

We need to argue that whatever our 'convergence' is, it's compatible with arguing that errors are small.

If we start a new Markov chain for each sample, we have independence: but if we keep running one Markov chain, we need to argue why samples from it are 'independent enough'.

There is a huge amount of literature on this, together with a lot of heuristic guesses—there are Markov chains which are widely used in practice even though we don't have (good) convergence guarantees, because we conjecture that the convergence is actually good even though we can't prove it. Markov chain Monte Carlo is used a lot in practice.

One thing you should be careful about, though, is 'tests for convergence' and 'autocorrelation'. It's common to simply run one Markov chain, wait until 'it has converged' and then take n 'sufficiently spaced samples' as our nearly-independent nearly-uniform samples. If you can rigorously prove these work, things are fine: if not, you may well be getting an estimate which is either far from being good (your chain hadn't really converged!) or where the probability bounds are not nearly as good as you think (the samples are very far from independent).

Many 'tests' used in practice are purely heuristic. In particular, consider the Markov chain of the infection process from Notes 4. We argued in Notes 4 Exercise 12 that for $\lambda > \rho$ this Markov chain will stay close to $\frac{\lambda}{\lambda-\rho}$ for an exponentially long time. This is more or less guaranteed to fool heuristic tests into saying that the chain has converged. But the stationary distribution is in fact (Exercise 5 from these notes) being at 0 with probability 1. This kind of behaviour can show up in other Markov chains: even ergodic chains can easily have big 'well-connected' regions of the state space, but with different regions being 'poorly connected', and standard tests may well say that 'the chain has converged' when it's actually likely only to have explored one of the several big regions; similarly, for this kind of chain you need to wait far longer between samples if you want 'nearly independent' samples.